Investing in Victory: The Roles of Team Salary and Performance Metrics in MLB Success (1985-2015)

Abstract

This study examines the impact of various team performance metrics on wins in Major League Baseball (MLB). Using data spanning 1985 to 2015, a fixed effect model with a log-log transformation is employed to investigate the relationship between performance metrics, finances, and team success. The results reveal that key predictors such as hits, home runs, walks, and runs against are consistently significant across both time periods. Furthermore, the elasticity analysis indicates a shift in the importance of some metrics over time.

Keywords: *Major League Baseball, Team Performance Metrics, Panel Data Analysis, Fixed Effects Model.*

Introduction

In Major League Baseball (MLB), wins are the ultimate measure of team success. However, the factors contributing to wins remain complex and multifaceted. Teams frequently evaluate performance using traditional and advanced metrics, but understanding how these variables translate into wins requires rigorous statistical analysis. This paper seeks to determine the elasticity of wins to various performance metrics, providing teams and managers with insights on how to best optimize their strategies.

Furthermore, by analyzing two distinct time periods, 1985-2000 and 2001-2015, this study aims to explore shifts in the strategic importance of these metrics over time. With the rise

of advanced analytics in baseball, particularly after the publication of Michael Lewis's Moneyball, the way teams evaluate talent and strategy has evolved significantly, making it crucial to understand the historical and contemporary factors that contribute to team success.

Objectives

This paper aims to identify and analyze the influence of specific performance metrics as well as team salary on the number of wins and compare how these relationships have evolved over two time periods: 1985-2000 and 2001-2015: A comparative approach will be used to identify shifts in strategic importance.

Literature Review

Financial incentives have frequently been employed as external motivators, with the belief that individuals are more inclined to exert themselves when monetary rewards are at stake. In Major League Baseball (MLB), the focus often lies on financial metrics such as player compensation and team revenue. However, understanding the significance of performance metrics is essential to comprehend how various performance metrics affect a team's success.

Research on the relationship between performance metrics and team success is relatively limited compared to financial metrics. Jacob Andrew Loree's 2016 paper, Determinants of Baseball Success: An Econometric Approach, provided a foundational understanding of the role non-financial metrics play in a team's total wins. Using performance metrics and econometric methods, Loree (2016) illustrated how various offensive, defensive, and financial indicators collectively shape a team's ability to win games. This literature review explores several studies, building on Loree's work.

Offensive and Defensive Performance

In an effort to identify which team statistics most significantly determine success, a regression analysis by Houser (2005) found that both offensive and defensive performance are crucial contributors to wins, aligning with Billy Beane's strategy for the 2002 Oakland A's. Similarly, Fullerton et al. (2014) analyzed the 2013 MLB season using regular-season wins as the dependent variable. Their study further emphasized the importance of strong team defense and offense, as well as pitching.

The question of whether defense or offense has a greater influence on wins was addressed by Pavitt (2011), who sought to find the balance between the two. Defense was defined as encompassing both fielding and pitching, while offense included only hitting, excluding baserunning statistics. The study concluded that defense and offense have equal impacts on team success, attributing this balance to the trade-off between the runs a team score and the runs it prevents the opposing team from scoring. However, Murray (2022) argued that limiting runs with pitching is better for teams than to scoring them. He conducted a side-by-side analysis of hitting and pitching statistics and found that pitching statistics consistently had lower p-values, concluding that they are more valuable to a team’s win total than scoring runs.

In a subsequent study, Pothula (2015) identified key performance metrics for team success using an Ordinary Least Squares (OLS) regression model to evaluate the relationship between offensive and defensive statistics and team wins. His findings highlighted the importance of Home Runs, Walks, Strikeouts, Throwing Errors, and Defensive Runs Saved in predicting wins. His study aligned with Loree (2016) results, demonstrating that offensive power

(Home Runs and Walks) and defensive efficiency (Fielding Errors and Defensive Runs Saved) are strong predictors of team success.

In exploring the relationship between various baseball statistics and team success, Loree (2016) used the Lahman Baseball Database to explore the relationship between various baseball statistics and team success. He identified critical factors influencing wins, such as the total number of runs conceded (Runs Against) and a team's ability to get on base via hits and walks. Offensive metrics like Total Hits, Doubles, Home Runs, and Walks significantly correlated with team wins, while defensive measures such as Errors and Fielding Percentage also affected wins.

A further examination of Success in MLB in Clark (2016) emphasized offensive and defensive statistics as predictors of success. His work identified Home Runs, Doubles, and Singles as critical offensive predictors of team wins, while pitching statistics like Earned Run Average (ERA), Walks Allowed, and Home Runs Allowed also significantly influenced win percentage. His results reinforced the importance of offensive metrics like Walks and Home Runs while highlighting defensive measures like Double Plays and Errors.

Financial Factors

In Major League Baseball (MLB), the absence of a strict salary cap has sparked ongoing discussions about the correlation between team payroll and on-field performance. Unlike counterparts such as the NFL and NBA, MLB operates under a luxury tax system, allowing teams to spend freely with penalties imposed for exceeding a predetermined threshold. This financial landscape has profound implications for competitive balance, as teams with larger budgets often wield a significant advantage in acquiring top talent.

However, the relationship between spending and success is complex, with numerous factors at play. Exploring the effects of salary on performance unveils interplay between financial resources and on-field results, offering insights into the dynamics of modern baseball. Loree (2016) explored the role of team salary in influencing team success. Although salary remained statistically significant, its impact on wins was limited. He found that the change in wins from the previous year significantly affected the following year's payroll, as successful teams could leverage their winning record to charge higher television fees and increase ticket prices.

The impact of total payroll on a team's winning percentage was investigated by Tartaro (2012) investigated the impact of total payroll on a team's winning percentage. Using stepwise regression, he revealed that payroll was not a significant predictor of winning percentage in the best model due to high correlations with other success factors. Variables such as Fan Attendance, Slugging Percentage, and Home Runs significantly contributed to payroll.

Comparing data across the four major North American leagues (MLB, NBA, NHL, and NFL) Shorin (2017) findings revealed that that higher payrolls are significantly associated with regular-season success across all leagues, though less so in the NFL due to a restrictive salary cap. In MLB, a higher payroll improves regular season winning percentage and postseason success due to the absence of a strict salary cap. Thus, salary remains a key focus in this analysis.

League Differences

It's essential to account for the differences between the American League (AL) and National League (NL) in Major League Baseball (MLB) to gain a comprehensive understanding

of player and team performances. The two leagues have distinct rules, including the designated hitter (DH) rule in the AL, which allows teams to use a specialized hitter instead of the pitcher for at-bats. This rule fundamentally alters offensive strategies and player utilization between the leagues Cronin (2016).

To account for such fundamental differences between the American and National Leagues, Loree (2016) included the AL Dummy variable in his analysis. Findings suggested that league membership plays a role in predicting team wins due to rule differences, such as the designated hitter rule in the American League. Clark (2016) also included league indicators and found that league differences affect team success, underscoring the need for league-specific strategies in baseball management.

According to the literature, offensive metrics like Hits, Doubles, Home Runs, and Walks are strong predictors of team wins. Defensive measures such as Errors and Fielding Percentage also impact wins, while financial factors (Team Salary) and league membership (AL Dummy) further differentiate team success levels. This paper uses similar metrics to build a predictive model for assessing and improving team wins, similar to Loree (2016), which used straight forward variables to provide insights into the dynamics of performance statistics and their effects on wins. In exploring this relationship, the null hypothesis posits that there is no significant relationship between performance statistics and wins in Major League Baseball (MLB), while the alternative hypothesis suggests that there is indeed a significant relationship worth considering.

Hypotheses

Null Hypothesis (H_0): Team salary has a weaker effect on team wins than performance metrics in Major League Baseball.

Alternative Hypothesis (H_1): Team salary has a stronger effect on team wins than performance metrics in Major League Baseball.

Methodology

This study utilizes a comprehensive dataset derived from the Lahman Baseball Database, covering the seasons from 1985 to 2015. It uses the data on pitching, hitting and fielding performance and other tables from 1871 through 2022, as recorded in the 2023 version of the database. This range was chosen as this was the earliest data available for team salary – a vital statistic for our estimation. The dataset includes yearly team performance metrics, enabling the examination of the salary and performance metrics in Major League Baseball (MLB).

Model Specification

To analyze the impact of team performance metrics on the number of wins, a panel data approach with fixed effects is employed to account for unobserved heterogeneity across teams and over time. The fixed effects model is particularly suitable for this analysis as it helps control for invariant characteristics within each team and year, allowing for a more accurate estimation of the effects of interest variables on team success. The estimation is carried out using the plm package in R, which is designed specifically for panel data analysis. This approach is validated by a Hausman test, which supports the use of fixed effects over random effects in this context, confirming that our model appropriately addresses the panel structure of the data.

The specification of the fixed effects model is as follows:

$$\text{Wins}_{it} = \alpha + \pi\text{Wins}_{it-1} + \beta\text{Batting}_{it} + \gamma\text{Pitching}_{it} + \delta\text{Fielding}_{it} + \theta\text{Salary}_{it} + \phi\text{AL} + p_t + t_i + \epsilon_{it}$$

Include controls for batting statistics (hits, doubles, triples, home runs, walks), pitching statistics (strikeouts, runs against), fielding statistics (errors, double plays, fielding percentage), and team salary. The team's league affiliation (American League or National League) is captured by a dummy variable. To ensure the robustness of the results, standard errors are clustered at the team level to correct for any serial correlation within teams over time. Furthermore, a series of diagnostic tests is conducted, including checks for multicollinearity, heteroskedasticity, and model stability over different time periods.

Data

This study utilizes a comprehensive dataset derived from the Lahman Baseball Database, covering the seasons from 1985 to 2015. It uses the data on pitching, hitting, and fielding performance and other tables from 1871 through 2022, as recorded in the 2023 version of the database. This range was chosen as this was the earliest data available for team salary – a vital statistic for our estimation. The dataset includes yearly team performance metrics, enabling the examination of the salary and performance metrics in MLB.

Table 1 demonstrates the summary statistics for the data. While the summary statistics align with the specifics of the game, Team Salary serves as an outlier due to factors such as inflation and media revenue Loree (2016). While the fixed effects model can mitigate some of these influences, they might not fully clarify the correlation between salary and performance. Table 2 further defines these variables. Table 2 gives the hypothesized relationships between wins and our independent variables of our data.

Most of these are intuitive to the sport of baseball but it is essential to crosscheck these against tour dataset. All variables, except for dummy variables, are log-transformed to capture the elasticity of team success relative to changes in performance metrics. This transformation addresses the non- linearity of effects and allows for a multiplicative interpretation of the regression coefficients.

Results

Table 3 shows us our regression results for the three models, the table explain that a one-unit change in the explanatory variable will lead to the coefficient change in Wins. It is important to note that Table 3 measures team salaries in thousands of dollars, since one dollar changes lead to extremely small changes in wins. Similarly, fielding percentage is measured as a one percent change (0.01 as opposed to 1) as it is measured between 0 and 1.

Fixed Effects Model

The fixed effects model provided a robust framework as shown in table 3, yielding an Adjusted R^2 of 0.857, which signifies a strong explanatory power regarding the variations in wins. Hits emerged as a particularly influential factor, with a coefficient of 1.029 ($p < 0.001$), suggesting that each additional hit is associated with approximately one additional win. This finding underscores the importance of offensive capabilities in baseball, which is in line with Clark (2016) emphasis on offensive being significant predictors of team wins.

Another key finding was the negative correlation between runs against and wins (-0.925, $p < 0.001$), confirming the pivotal role of strong pitching and defense in limiting opponents' scores to enhance win probabilities. Furthermore, Team salary has some level of

impact, it does not operate in isolation and its effects on team success are relatively constrained when considering the incremental financial investments. The results align with Loree's (2016) findings, where salary remained statistically significant but with limited impact on wins. This supports the notion that other factors, such as team dynamics and strategic plays (like walks and home runs), may have a more pronounced influence on a team's success than salary alone.

Linear Panel Model (1985-2000)

Transitioning to the linear panel model covered data from the years 1985 to 2000 and controlled for year and team fixed effects, lead to some shifts in the significance and magnitude of coefficients. Hits and walks continued to show a positive and significant relationship with wins, although the coefficients were slightly reduced compared to the fixed effects model. The negative impact of runs allowed remained robust and significant, reinforcing the importance of a strong defense as noted in prior studies like those by Houser (2005) and Fullerton et al. (2014). The coefficient for errors increased (coefficient = 0.454, $p < 0.01$), and fielding percentage had an even larger coefficient (48.765, $p < 0.01$) than in the fixed effects model. This suggests a changing dynamic in the game during this period, where fielding became even more crucial to team success.

However, as hypothesized, the coefficient for team salary was small and non-significant. This indicates that financial expenditures on player salaries did not translate directly into wins, a finding that reflects the complexity of team dynamics and the multifaceted nature of baseball success as observed by Tartaro (2012)

Linear Panel Model (2001-2015)

The model covering the years 2001-2015 showed an Adjusted R^2 of 0.826, slightly lower than the earlier periods, perhaps due to evolving strategies that are not fully captured by the traditional metrics used. The significant reduction in the coefficient for stolen bases to a non-significant level (0.012) underscores a strategic departure from speed-focused tactics to perhaps more power-centric approaches, as evidenced by the increased significance and magnitude of home runs (0.235, $p < 0.001$).

Conversely, the importance of homeruns increased (coefficient = 0.235, $p < 0.01$), indicating a potential shift towards power hitting in modern baseball strategies. Interestingly, during this period, the significance of fielding percentage drastically reduced to an insignificant level with a negligible coefficient (0.682), which might suggest a general improvement in fielding across the league, reducing its differentiation as a competitive advantage.

Analysis

The results from the three models reveal a clear evolution in the strategic elements influencing Major League Baseball wins. Early on, traditional skills like hits and stolen bases played dominant roles, but over time, the emphasis appears to have shifted towards power-hitting as indicated by the increasing impact of home runs. The diminishing role of fielding percentage and stolen bases in the more recent model suggests a potential shift in the game's strategic priorities or perhaps a leveling in the quality of fielding across teams.

Errors and their surprising positive correlation with wins in earlier models might be indicative of an aggressive style of play that, while occasionally leading to mistakes, also results in more dynamic and potentially scoring plays. This aspect warrants further investigation to understand the underlying behaviors fully.

Finally, Figure. 1 illustrates the variation in the number of wins across the American League (AL) and National League (NL) over a thirty-year period. The median wins are comparable between the two leagues, yet the American League shows a notably lower minimum, suggesting occasional seasons of significantly poorer performance compared to the National League. Both leagues exhibit a similar range of outcomes in terms of upper quartile and maximum values, which indicates that the top-performing teams in both leagues achieve a comparable number of wins.

The presence of outliers in the American League, particularly at the lower end, might reflect specific years where certain AL teams underperformed dramatically compared to their NL counterparts. This visualization effectively encapsulates the competitive balance between the leagues, which possibly suggest that factors such as the designated hitter rule's do have an impact on game outcomes.

Conclusion

The comprehensive analysis across multiple models and time periods consistently demonstrates that performance metrics have a more substantial impact on MLB team wins than team salary. The statistically insignificant impact of salary on wins aligns with the Null Hypothesis (H_0), which posits that team salary has a smaller impact compared to performance metrics. These findings highlight the relative importance of strategic gameplay and player performance over financial spending alone.

This study underscores the dynamic nature of baseball, with clear shifts in the strategies that correlate with winning over the past few decades. While certain fundamentals like hitting remain critical, the importance of power-hitting and the reduced emphasis on speed and fielding precision reflect broader changes in the game. Teams looking to capitalize on these insights must consider evolving their strategies to align with these trends, focusing on developing power hitters and perhaps investing more in pitching to reduce runs against, which consistently showed a strong negative correlation with wins.

Overall, the results from offer a comprehensive understanding of how various aspects of the game contribute to team success and help divert attention from the absence of salary caps. These findings could be pivotal for team managers and coaches as they develop strategies to maximize their chances of winning. Future research could benefit from integrating advanced analytics to explore these trends. While team salary in MLB has a role in shaping team success, its influence is neither direct nor overwhelmingly significant. These findings indicate successful teams are likely leveraging financial resources not just in acquiring talent, but more importantly, in enhancing team performance through strategic decisions on player selection and game tactics.

Table 1. Summary Statistics

	Minimum	1st Quartile	Median	3rd Quartile	Maximum	Mean
Wins	43.00	71.75	80.00	89.00	116.00	79.91
Hits	1139	1379	1436	1495	1684	1431
Doubles	159.0	255.0	276.0	297.0	376.0	275.3
Triples	11.00	24.00	30.00	37.00	61.00	31.03
Homeruns	58.0	129.0	156.0	181.0	264.0	157.1
Walks	319.0	475.0	523.5	576.0	775.0	523.5
Strikeouts	568	926	1036	1133	1535	1036
Stolen Bases	25.0	78.0	101.0	127.0	314.0	104.9
Runs Against	448.0	665.8	729.0	793.2	1103.0	730.2
Completed Games	0.00	4.00	7.00	13.00	47.00	9.72
Errors	54.0	95.0	108.0	123.0	181.0	109.6
Double Plays	82.0	135.0	148.0	161.0	204.0	147.7
Fielding Percentage	0.9710	0.9800	0.9820	0.9840	0.9910	0.9819
Al Dummy	0.0000	0.0000	0.5079	1.0000	1.0000	0.5079
Team Salary	880000	24375500	48789283	81667000	231978856	57847972

Table 2. Hypothesized Variable relationships to Wins.

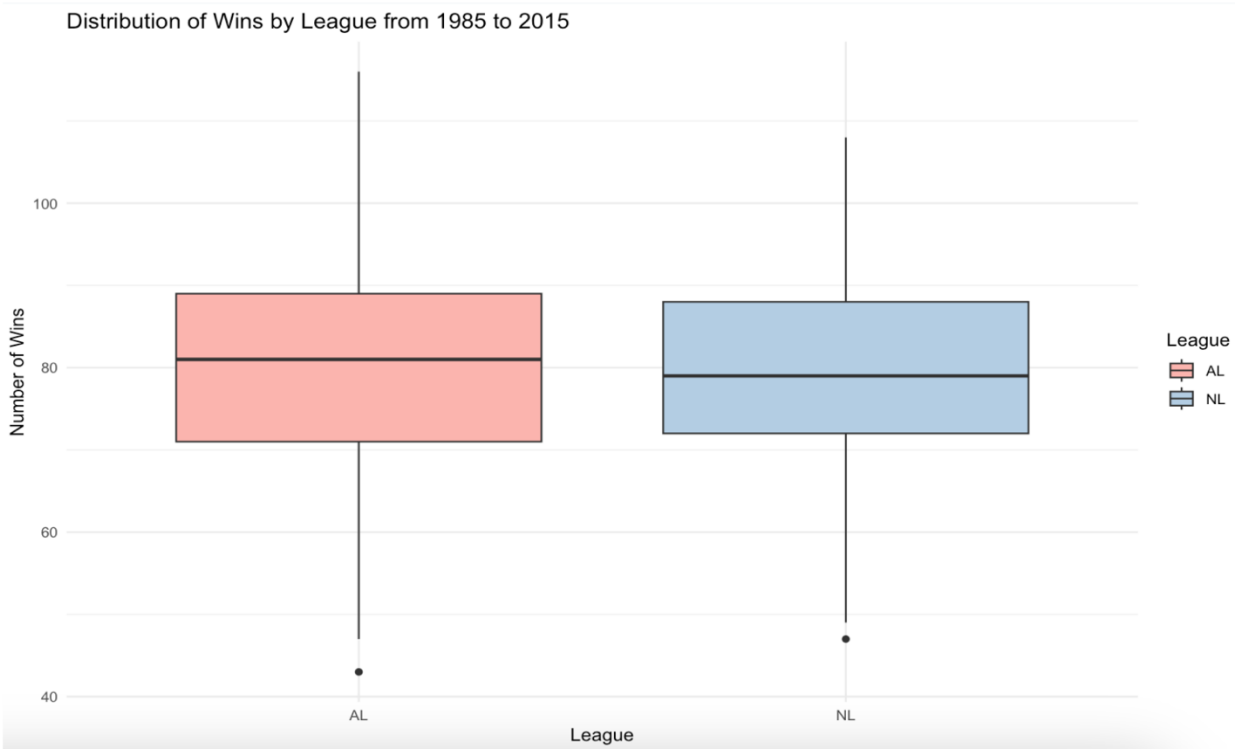
	Effect on Wins
Hits	Positive
Doubles	Positive
Triples	Positive
Homeruns	Positive
Walks	Positive
Strikeouts	Negative
Stolen Bases	Positive
Runs Against	Negative
Completed Games	Positive
Errors	Negative
Double Plays	Positive
Fielding Percentage	Positive
Al Dummy	Positive

Table. 3 Unit Change Regressions.

	Fixed Effects Model	Linear Panel (1985-2000)	Linear Panel (2001-2015)
Hits	1.029***	0.877***	0.752***
	(0.051)	(0.084)	(0.092)
Doubles	0.052*	0.083*	0.124**
	(0.024)	(0.041)	(0.045)
Triples	0.014	0.025	0.014
	(0.008)	(0.013)	(0.012)
Homeruns	0.188***	0.198***	0.235***
	(0.011)	(0.018)	(0.019)
Walks	0.250***	0.269***	0.210***
	(0.017)	(0.029)	(0.028)
Team Salary	0.001	0.002	0.005
	(0.004)	(0.006)	(0.010)
Strikeouts	-0.009	-0.019	-0.068*
	(0.019)	(0.035)	(0.034)
Stolen Bases	0.034***	0.042***	0.012
	(0.006)	(0.011)	(0.010)
Runs Against	-0.925***	-0.908***	-0.946***
	(0.021)	(0.036)	(0.033)
Completed Games	0.000	-0.001	-0.001
	(0.004)	(0.007)	(0.006)
Errors	0.337***	0.454***	-0.006
	(0.048)	(0.075)	(0.095)
Double Plays	-0.032	-0.028	-0.060*
	(0.019)	(0.028)	(0.028)
Fielding Percentage	39.229***	48.765***	0.682
	(5.528)	(8.031)	(11.684)
Al Dummy	-0.011*	0.077*	0.077*
	(0.005)	(0.039)	(0.039)
Observations	888	438	450
Adjusted R^2	0.857	0.874	0.826
Year and Team Fixed Effects	No	Yes	Yes

Standard errors are in parenthesis. ***, **, * indicate 0.1, 1, 5% significance respectively.

Figure. 1 Distribution of Wins by League from 1985 to 2015



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